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Mangrove tree height growth monitoring from multi-temporal UAV-LiDAR

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ABSTRACT

Quantifying the mangrove height growth rate is important for understanding mangrove growth as well as its coupling with the other ecosystems and the climate. However, research on this topic has been limited due to the lack of multi-temporal data on mangrove tree height (TH) and methods for tree-level analysis. This study investigates the use of multi-year UAV-LiDAR data for mangrove TH growth characterization. We evaluated the UAV-LiDAR data by comparing it with field measured THs, observing a deviation ranging between −9 and +11 cm. Notably, the deviation was minimal at only 1 cm when comparing datasets collected using the same LiDAR sensor, indicating the validity of adopting multi-year UAV-LiDAR data for mangrove TH growth analysis. Methodologically, we assessed a tree-based approach and a grid-based approach. The tree-based approach is considered more feasible as it mirrors field measurements, allowing for direct comparisons with field surveys. In addition, statistically significant differences were revealed in TH growth rates among mangrove species and clumpiness groups ($p < 0.001$), while the invasion of *Spartina alterniflora* did not significantly affect the growth rates. This paper presents the inaugural effort in mangrove TH growth quantification using multi-temporal UAV-LiDAR data, showcasing the potential of this quickly emerging remote sensing technique in individual mangrove change analysis. We proposed a recommended framework for UAV-LiDAR-based TH growth rate measurement. The limitations and uncertainties embedded in the data sources, methodology, and results interpretation were also discussed.

1. Introduction

Mangrove forests, predominantly found along tropical and subtropical coastlines (Giri et al., 2011; Kuenzer et al., 2011), play a crucial role as significant blue carbon reservoirs (Pendleton et al., 2012; Wang et al., 2019). Their sustained and healthy growth is vital for maintaining their ecosystem services in regulating (Alongi, 2008; Donato et al., 2011; Pendleton et al., 2012), provisioning (Cannicci et al., 2008; Duke, 2017), supporting (Kathiresan and Bingham, 2001; Nagelkerken et al., 2008), and cultural (Mitra, 2020; Rivera-Monroy et al., 2017) aspects. To measure the growth of mangroves is especially important in understanding their contribution to achieving carbon neutrality and alleviating global climate change (Song et al., 2023; Zhu and Yan, 2022). Thus, quantifying the mangrove growth rate is imperative as it not only keeps us informed of the mangroves' proper functioning, but also signifies the degree to which coastal vegetation is impacted by and has adapted to sea level rise induced by the escalating climate change (Chowdhury et al., 2016; Raha et al., 2012).

Mangrove growth occurs in the three-dimensional (3D) world, at both horizontal and vertical dimensions. While the horizontal growth of mangroves has been well-documented through various mapping studies (Jia et al., 2018), the vertical growth (i.e., tree height growth) has received relatively little attention. This research gap can be attributed to the low accessibility of mangrove trees. Because mangrove trees are in the densely forested intertidal zones, traditional on-site measurements of tree heights (THs) pose significant challenge and high uncertainty (Lagomasino et al., 2016). Moreover, to understand how the TH growth correlates with different environmental factors is even more difficult.

To alleviate these problems, remote sensing provides a viable avenue for measuring mangrove THs at spatially continuous and temporally frequent extents (Simard et al., 2006; Wang et al., 2004b; Yin and Wang, 2019). Light detection and ranging (LiDAR), in particular, directly characterizes 3D structures and has proven effective in accurately measuring mangrove THs (Chadwick, 2011; Lagomasino et al., 2016;

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Simard et al., 2006).

Despite the handful attempts striving to quantify mangrove THs at single time points with LiDAR, no studies have yet utilized multi-temporal LiDAR to track mangrove TH growth (Wang et al., 2019). This lack of research can be attributed to two primary factors: accuracy and cost. First, the accuracy of LiDAR-based mangrove TH measurements often falls short of the precision needed for the growth rate quantification. While mangrove growth rates as low as 5 cm per year have been reported (Wannasiri et al., 2013), the error of TH measurements using LiDAR data (airborne) often exceeds 40 cm (Jakubowski et al., 2013; Wannasiri et al., 2013). Second, the cost for obtaining multi-temporal LiDAR is high. The cost to obtain airborne LiDAR data has been reported to be over a thousand dollars per hectare (Zhao et al., 2018).

Unmanned aerial vehicles (UAVs) have emerged as a flexible and cost-effective alternative to address the aforementioned challenges in accuracy and cost (Tian et al., 2017; Yin and Wang, 2019). The latest professional UAV (e.g., DJI M300RTK) together with a compatible LiDAR sensor (e.g., DJI Zenmuse L1) can be purchased for approximately $20,000 (US dollars), with necessary preprocessing software included. Therefore, now with the improved ranging accuracy of LiDAR sensors and enhanced spatial resolution of UAV data acquisition, it is the perfect time to introduce multi-temporal LiDAR change analysis to the field of mangrove TH growth monitoring (Wang et al., 2019). Nevertheless, research into change analysis using multi-temporal UAV-LiDAR data, particularly in the context of mangrove TH growth characterization, remains unexplored. The following two challenges are envisaged.

(1) What is a feasible approach to derive TH growth from UAV-LiDAR data?

Relatively low density multi-temporal airborne LiDAR has been utilized to quantify the growth of trees larger than mangroves, with the prerequisite that the uncertainty of growth estimation is within the growth range. The usefulness of bi-temporal airborne LiDAR for tree growth estimation in the boreal forest has been demonstrated in southern Finland two decades ago (Yu et al., 2004). Success in estimating individual tree growth was achieved later for Scots pine trees from 1998 to 2003, with good agreement to field observations ($R^2 = 0.68$) (Yu et al., 2006). Besides the change detection between two timestamps, recent works used multi-temporal datasets to monitor TH changes in both broadleaf (Xiao et al., 2016) and coniferous plantations (Ma et al., 2018; Zhao et al., 2018). Zhao et al. (2018) found an average growth rate of 0.62 m/year with 0.24 m/year standard error, indicating that multi-temporal LiDAR could effectively measure TH growth over six-month intervals.

Two types of approaches have been demonstrated successful in quantifying TH growth with multi-temporal LiDAR: grid-based and tree-based. The grid-based approach first generates a canopy height model (CHM) to represent the canopy height for each pixel or grid, and then calculates the difference between multi-temporal CHMs directly (Hopkinson et al., 2008; Yu et al., 2008). This approach is relatively easy to implement. However, the estimated growth includes height difference at both the treetops and other canopy parts, which differs from the normally expected TH growth (i.e., the increase of treetop height). The tree-based approach, in contrast, derives TH growth metrics that well corresponds to field measurements and intuitive understanding (Yu et al., 2006). Such methods first detect individual trees, then measure the treetop heights, and finally compare the individual THs over time (Ma et al., 2018; Zhao et al., 2018). However, an obstacle hindering the wide use of this approach is the difficulties associated with individual tree delineation (Yin and Wang, 2019). To select the optimal technique for mangrove TH growth characterization, a comparison of these two approaches is needed.

(2) How do mangrove growth rates vary across different species, clumping densities, and invasion statuses?

The mangrove growth rates vary due to different factors, the understanding of which requires quantifying the growth rate of different mangrove groups. Among all the factors that may affect the growth rate, some are most likely to be detected from remote sensing, including species, clumping density, and invasion. In terms of species, growth rates of 5 cm to 1.18 m per year have been reported for different mangrove species and sites (Ha et al., 2003; Kairo, 1995; Wannasiri et al., 2013). Yet the growth rate for many species and sites remain unavailable, primarily due to the absence of an efficient approach to measure such information. In terms of clumping density, mangroves often present high clumpiness with adjacent tree crowns tightly interconnected. An assumption has been raised that species growing around large gaps grow faster than those with small gaps (Imai et al., 2009), yet this hypothesis needs further validation with concrete mangrove growth data. In terms of invasion, the invasive *Spartina alterniflora* has been found to alter the physiological factors of mangroves such as light use efficiency and leaf area (Yang et al., 2018). It is fair to assume that the growth rate may also be affected because they compete for nutrients, water, and light. Yet no studies have examined the impact of species invasion on mangrove growth. To fill the knowledge gaps regarding the variation in mangrove growth rate, we propose to separately gauge growth rates for different mangrove groups (e.g., species) using multi-temporal UAV-LiDAR.

Hence, to investigate the two above-mentioned issues, we propose to characterize mangrove TH growth with multi-temporal UAV-LiDAR data. Specifically, we will answer two key questions: (1) Which is a more feasible approach for deriving TH growth from UAV-LiDAR data, the grid-based or the tree-based approach? and (2) What are the growth rates for different mangrove groups (i.e., species, clumpiness, and invasion) manifested in multi-temporal UAV-LiDAR? This is the first study to monitor mangrove TH dynamics using multi-temporal UAV-LiDAR, and also the first study to use multi-temporal UAV-LiDAR to monitor TH growth in all forest types. We anticipate that the findings of this work will provide valuable insights into mangrove growth monitoring, as well as enhance the efficiency of tree growth monitoring in forests globally.

The overall experimental design is presented in section 2, concerning study area, data collection, multi-temporal change analysis, and the three factors affecting mangrove TH growth rate (species, clumpiness, invasion). Section 3 lists the experiment results regarding the two research questions posed. Section 4 discusses the significance and limitations of this study. Finally, the major conclusions are summarized in section 5.

2. Materials and methods

To solve the two research questions, we conducted experiments following Fig. 1. Firstly, both field data and UAV-LiDAR data were collected in four consecutive years (section 2.1). The UAV-LiDAR data were then co-registered and used in a four-year multi-temporal growth analysis to characterize the mangrove TH growth. Aiming at answering the first research question, i.e., finding an approach to measure the mangrove growth accurately and efficiently, the analysis was conducted using both grid-based and tree-based approaches (section 2.2). In the grid-based approach (section 2.2.1), the CHMs generated from the four UAV-LiDAR datasets were directly used for the multi-temporal analysis. In the tree-based approach (section 2.2.2), the four UAV-LiDAR datasets were separately used for individual tree detection and crown delineation (ITDD) to achieve TH of individual trees. These trees were then matched across different years to provide input for the multi-temporal growth analysis. Finally, the selected approach was adopted in the next step to analyze the impact of different factors (species, clumpiness, invasion) on mangrove growth rates (section 2.3), which would provide answers to the second research question.

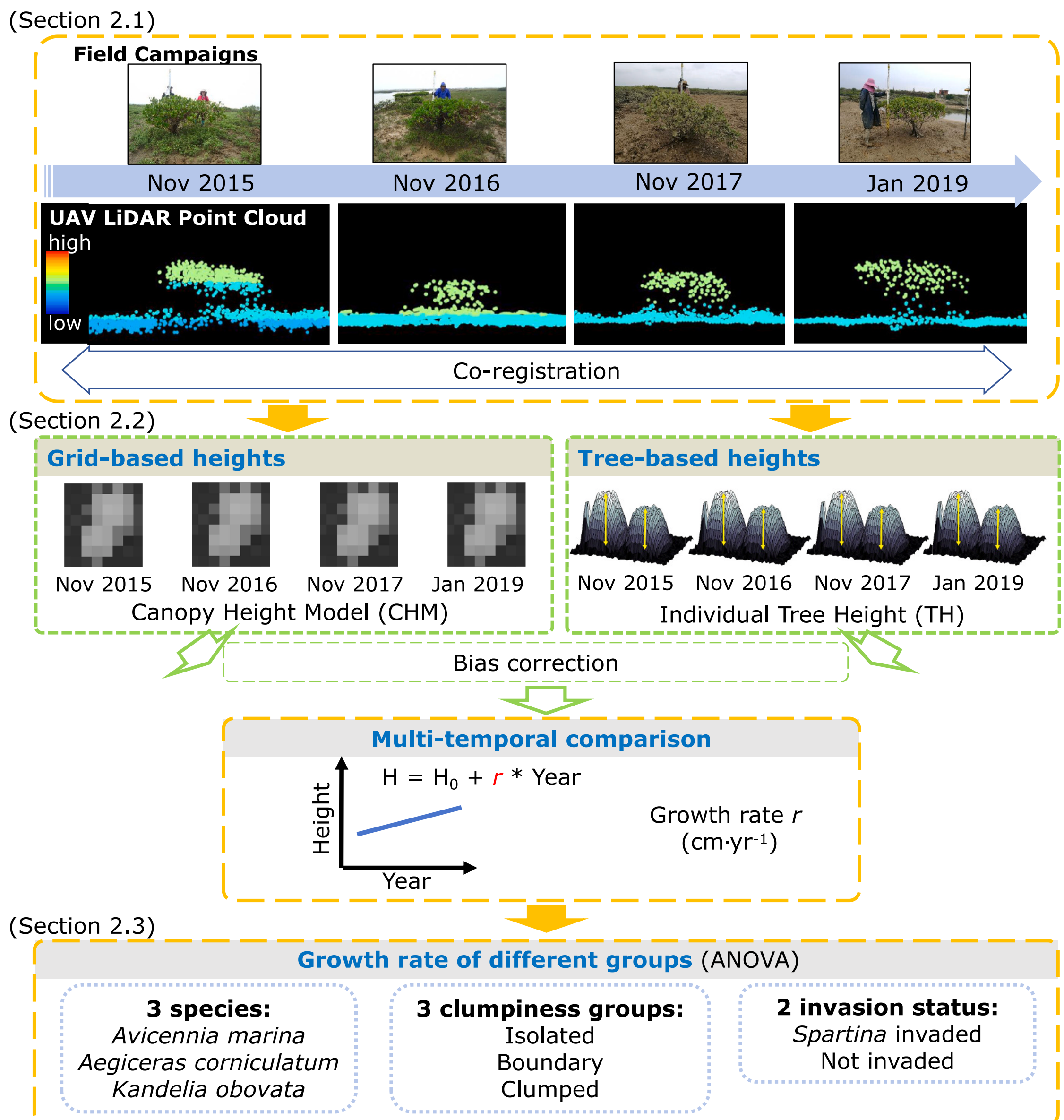

Fig. 1. Flowchart of this study. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2.1. Data collection and preprocessing

2.1.1. Study area

We surveyed mangroves along the Dandou Sea coast (21°35'30" N, 109°40'0" E) in Beihai City, Guangxi Zhuang Autonomous Region, one of the largest mangrove habitats in China (Fig. 2) (Jia et al., 2014; Li and Lee, 1997). The study area is at the north landward fringe of the Shankou National Mangrove Nature Reserve (established since 1990). This reserve is a vital habitat for 15 mangrove species, along with a wide variety of rare plants and animals (United Nations Educational, Scientific and Cultural Organization, 2007; Ministry of Environment Protection of the People's Republic of China, 2016). This area experiences a subtropical oceanic monsoonal climate. The maximum and minimum monthly average temperature in the Beihai City from 2000 to 2022 are 29.4 °C in July and 15.3 °C in January respectively. The average annual precipitation is 1806 mm (meteorological data accessed through http://www.meteomanz.com/). The dominant tidal pattern is regular diurnal, while the tidal current pattern is mixed mainly diurnal (Gao et al., 2019).

Dominant mangrove species in the study area are *Avicennia marina*, *Aegiceras corniculatum* and *Kandelia obovata*. Another two mangrove species *Rhizophora stylosa* and *Bruguiera gymnorrhiza* also exist here at relatively smaller amounts. Most mangroves in this area are lower than 3 m, typical of the short mangrove varieties worldwide (Amjad et al., 2007; Li, 2003, 2004; Simard et al., 2006). The vertical structure in the study area is simple, with no grown vegetation under mangrove crowns (Yin and Wang, 2019). An invasive species *Spartina alterniflora* has been rapidly encroaching upon the mangroves in the study area, since its introduction to China in 1979 as a coastal erosion control measure (Tian et al., 2020). The vertical accretion rate of the substrate elevation in our study area was estimated to be 0.1 cm•yr^{-1} (Li et al., 2015). As 0.1 cm•yr^{-1} is much lower than the growth rate of mangrove trees and the measurement precision, we omitted the substrate accretion in this study.

We did not find field research specifically detailing the age of mangroves in the Dandou Sea area. Considering that the growth rate varies depending on the age and growth stage (Lin, 1997), we looked for auxiliary sources to help infer the mangrove age in our study site. It is described in the literature that the mangroves in this region (Dandou Sea area) have undergone repeated felling (Miao et al., 2007). About 55% of mangroves in China were lost due to facility construction between 1980

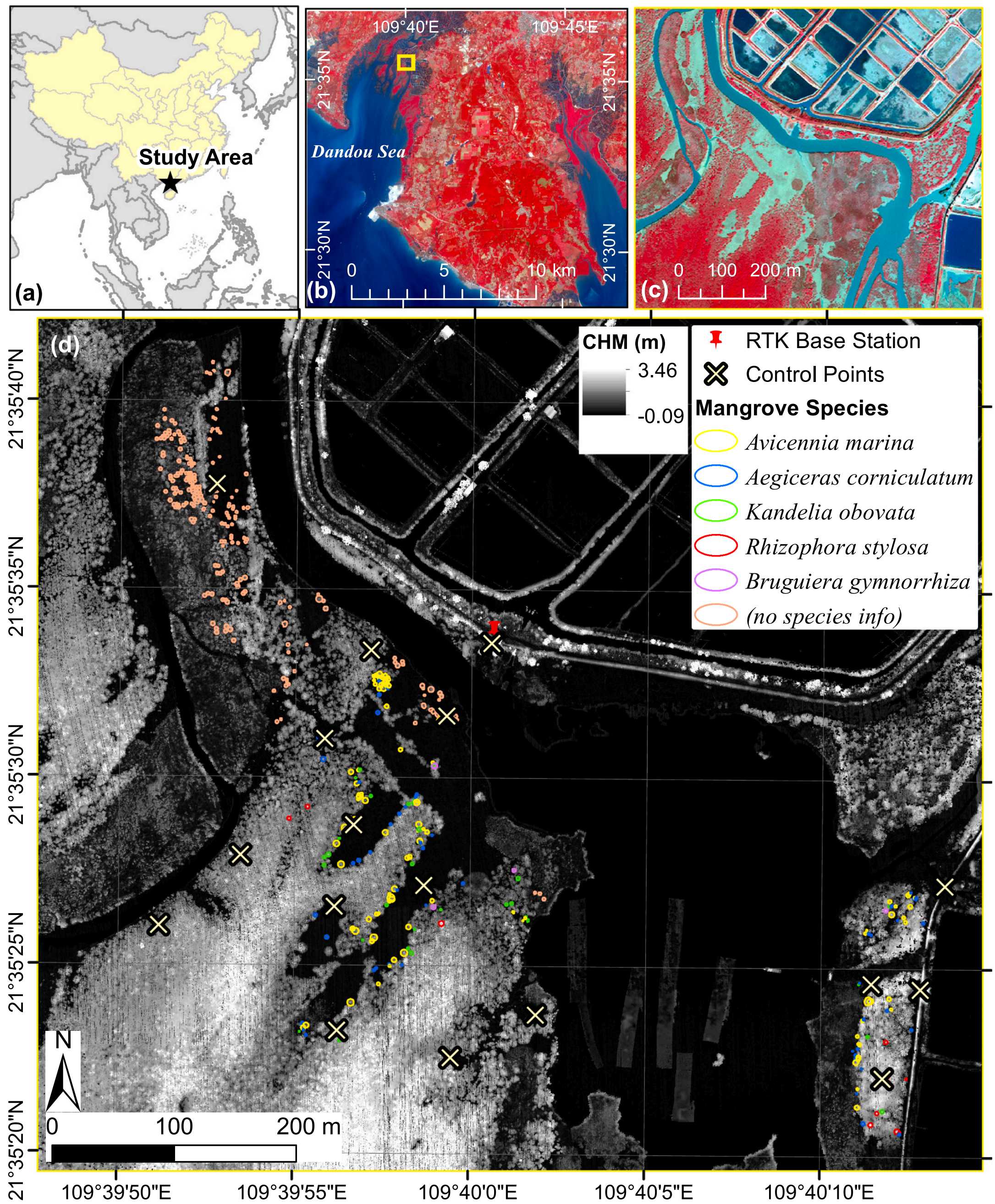

Fig. 2. Study area. (a) Location of the study area, marked by the star. (b) Multispectral false color composite of the Shankou National Mangrove Nature Reserve (Image source: Sentinel-2A on Jan 23, 2019). The yellow box shows the extent of our study area. (c) Multispectral false color composite of the study area (Image source: WorldView-3 on Oct 19, 2013). (d) CHM of the study area in Jan 2019, circles marking sampled individual mangroves. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and 2000, while the mangrove extent has been recovering after 2000 thanks to conservation actions (Chen, 2021). By inspecting Google Earth images, high spatial resolution (0.5–10 m) satellite images, and medium resolution (30 m) Landsat images, we found that many of the sampled trees did not exist before 1989, which is consistent with Jia et al. (2018). Heavy human activities such as fish/shrimp pond building were also observed in the area before 2000, leading to significant mangrove habitat loss, which indicates that some of the mangrove trees can be grown post-2000. In addition, considering that our dataset includes both small (below 1 m) and large (over 3 m) mangrove trees, we infer that our study encompasses both young (below 10 year) and older (over 20 year) mangrove trees.

2.1.2. Field data

We conducted field campaigns in 2015, 2016, 2017, and 2019. During each campaign, we remeasured the previously measured trees and added additional samples. Each tree was marked by attaching a piece of red cloth with a unique tree identifier written on it. Altogether, we recorded the location and TH for 225 sampled trees. Individual mangrove samples were selected randomly in the field. For each sampled mangrove tree, we measured the center location using a real-time kinetic global positioning system (RTK-GPS) with centimeter-level accuracy. However, due to obstruction by the branches, the GPS could not always be placed at the center of the crowns. Therefore, we manually edited the recorded locations to align them with the crown centers. Each TH was determined with a leveling staff. Crown width was measured with a tape ruler. Reference crown boundaries were created as circular buffers using the crown centers and radii.

Given the high density of trees, we manually added samples to make our findings more representative. The crown boundaries of these additional samples were visually interpreted from the UAV data. As a result, we had 454 tree samples (Fig. 2d). Note that these interpreted tree crowns were only used for checking the tree locations but not for TH accuracy assessment because their THs were not measured.

2.1.3. UAV-LiDAR data

We acquired UAV-LiDAR data in the winter of four consecutive years, namely Nov 2015, Nov 2016, Nov 2017, and Jan 2019. The flights were conducted in the winter time because the UAVs need to fly under dry skies and storms are least possible to occur in the study area from October to February as indicated by meteorological data (accessed through http://www.meteomanz.com/). Each UAV-LiDAR data collection coincided with a field campaign except for the Nov 2015 survey. Due to scheduling conflicts, we could not measure THs in the field in Nov 2015. Luckily, we had field data in May 2015 and Jun 2016. Hence, for the Nov 2015 UAV data, we used the average of field-measured heights from May 2015 and Jun 2016 as reference THs.

During each UAV data collection, RTK-GPS was used to optimize the positioning accuracy and minimize the spatial displacement of the LiDAR data. The base station for the RTK-GPS was placed on the same rooftop location in the Nov 2015 and Nov 2016 surveys, which was marked and reused in the four years (as marked by the red pin in Fig. 2d). Later in the Nov 2017 and Jan 2019 surveys, the network RTK-GPS service was used. More details about the scanners and data products are outlined in Table 1. Each dataset was processed by the vendor to separate ground points and non-ground points based on the geometric differences in the LiDAR point cloud (Zhao et al., 2016). The digital surface model (DSM), the value of each pixel representing the land surface elevation, was generated by averaging and interpolating the last return LiDAR points. The digital terrain model (DTM), the value of each pixel representing the ground elevation, was generated by averaging and interpolating the ground points. The difference between the ground elevation and the canopy surface elevation was then calculated to represent the canopy height (Yin and Wang, 2019).

A prerequisite for tree growth analysis using multi-temporal LiDAR is the accurate co-registration of LiDAR datasets. Although RTK-GPS was enforced for LiDAR data collection, a displacement among the four datasets was still observed (Fig. 3a). To address ths, we tried out various registration algorithms, including the most used point cloud registration method iterative closest point (ICP) algorithm (Yu et al., 2006; Marinelli, 2019) and the popular image registration method polynomial georeferencing (Duncanson and Dubayah, 2018). Ultimately, we decided to register the datasets by shifting the CHMs till the highest correlation was reached. The dataset collected in Jan 2019 served as the fixed reference and the other three were moved towards it. We identified 18 control points in the study area (Fig. 2d) to quantitatively assess the relative positional accuracy of the LiDAR data (Fig. 3b). Following the process, the average spatial displacements in the 2015, 2016, and 2017 datasets relative to the 2019 LiDAR data were significantly reduced, from initial displacements of 3.05 m, 2.93 m, and 1.47 m to just 0.68 m, 0.39 m, and 0.58 m, respectively.

Table 1
UAV data collection summary.*

	Time	Nov 2015	Nov 2016	Nov 2017	Jan 2019
Scanner specifications	LiDAR scanner	Velodyne HDL-32E	SKY-Lark AP-0300	Riegl VUX-1LR	Riegl VUX-1LR
	Wavelength (nm)	903	1550	1550	1550
	Pulse rate (kHz)	700	50–600	< 820	< 820
	Range Accuracy (cm)	2	1.5	1.5	1.5
Flight specifications	Flight altitude (m)	40	–	160	150
	Flight speed (m/s)	4.8	–	7	6
Data summary	Number of returns	1	4	6	7
	Point density (pt/m^2)	91	227	89	57
	Auxiliary UAV photo	Yes	Yes	Yes	Yes
	Corresponding field campaign	May 2015, Jun 2016	Nov 2016	Nov 2017	Jan 2019

* The flight specifications for the Nov 2016 data collection were not retrievable. Because these details are not used in the TH growth analysis, their absense is deemed non-critical for this paper.

2.2. Multi-temporal change analysis

To identify the optimal approach for deriving mangrove TH growth with UAV-LiDAR data, both grid- and tree-based approaches were implemented in this study. The accuracies were assessed separately and compared subsequently.

2.2.1. Grid-based approach

In the grid-based approach, canopy height was determined by the pixel value in CHMs. TH growth was estimated by computing pixelwise CHM differences. To alleviate the error caused by positional displacement, the CHM was generated at 1 m spatial resolution. We manually delineated a mask for the mangrove area so that water, mud flats, and grass (e.g., *Spartina alterniflora*) were excluded in the growth analysis.

The growth trajectory of each pixel was visualized by using the time-series CHM data from Nov 2015 to Jan 2019 to show the overall distribution of growth across the study area. Furthermore, the average height in each year was calculated to depict the average growth trajectory of the mangroves. Via building a linear regression model with average height as the dependent variable and year as the independent variable, we represented the average annual growth rate (in cm•yr^{-1}) by the slope.

2.2.2. Tree-based approach

(1) Individual tree delineation

To delineate individual mangrove tree crowns is a crucial step for tree-based growth analysis. There are different ITDD methods such as seeded region growing (SRG) (Hyyppä et al., 2001; Zhen et al., 2015), marker-controlled watershed segmentation (MCWS) (Wang et al., 2004a; Yin and Wang, 2019), and deep learning algorithms. MCWS was found to be more accurate than SRG in our study area (Yin and Wang, 2019). Lately, a deep learning approach mask region-based convolutional neural network (Mask R-CNN) has achieved high accuracy for ITDD (Hao et al., 2021). In addition, the you only look once version 7 (YOLOv7) algorithm has proved more accurate and efficient than Mask R-CNN (Wang et al., 2022) but has not been used for ITDD. Therefore, we compared the three state-of-the-art methods (MCWS, Mask R-CNN, and YOLOv7) and used the most accurate one to delineate individual mangrove tree crowns from each of the LiDAR datasets and measured their TH. Details of the three algorithms are as follows.

MCWS algorithm (Fig. 4): Firstly, we created the CHM at 0.25 m spatial resolution because 0.25 m was found to be the optimal resolution for mangrove ITDD in our study area (Yin and Wang, 2019). A finer spatial resolution was not used to avoid the negative impact of excessive spatial details (Treitz and Howarth, 2000) and empty pixels with no LiDAR points (Yin and Wang, 2019). Secondly, considering that taller trees usually have wider crowns, we built a linear regression model between TH and crown diameter (CD) using the field measured data, to

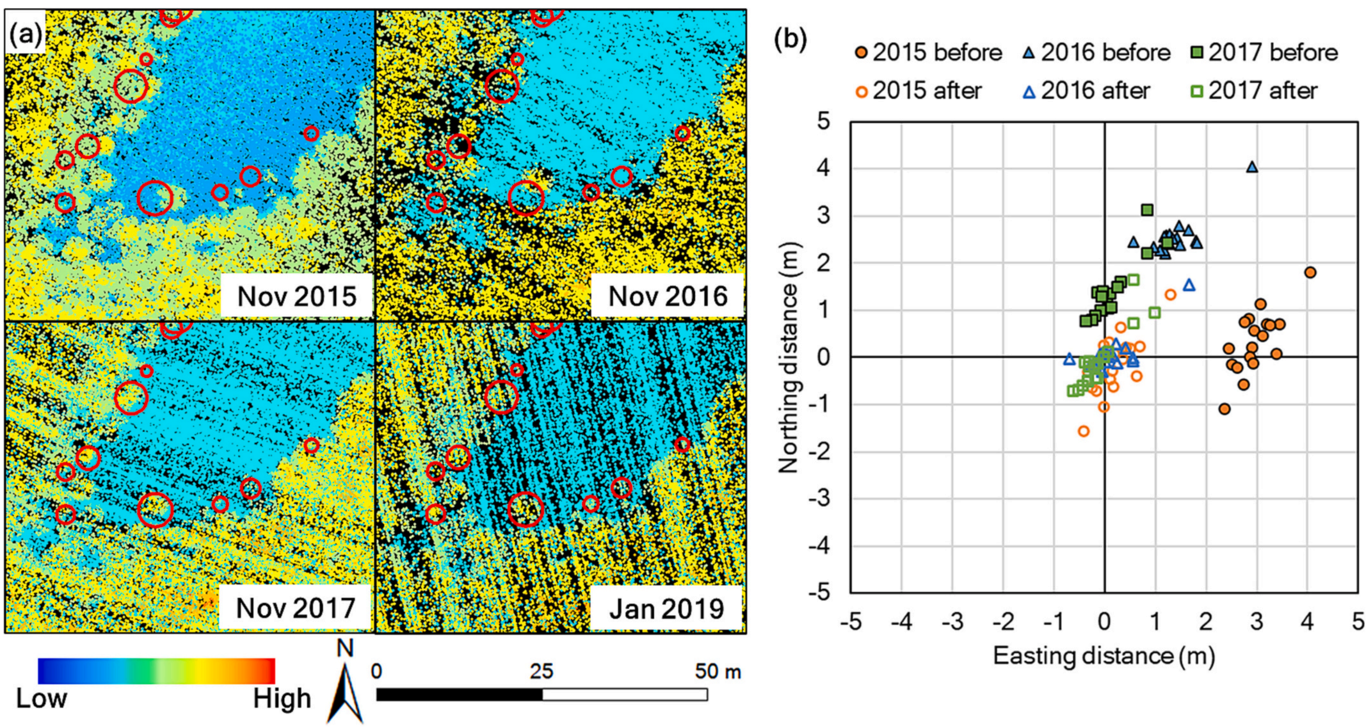

Fig. 3. Spatial displacements in the four UAV-LiDAR datasets. (a) UAV-LiDAR data, red circles marking the sampled mangrove tree crowns. (b) Distance of the control points in the CHMs of 2015, 2016, and 2017 from the 2019 CHM, before and after co-registration. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

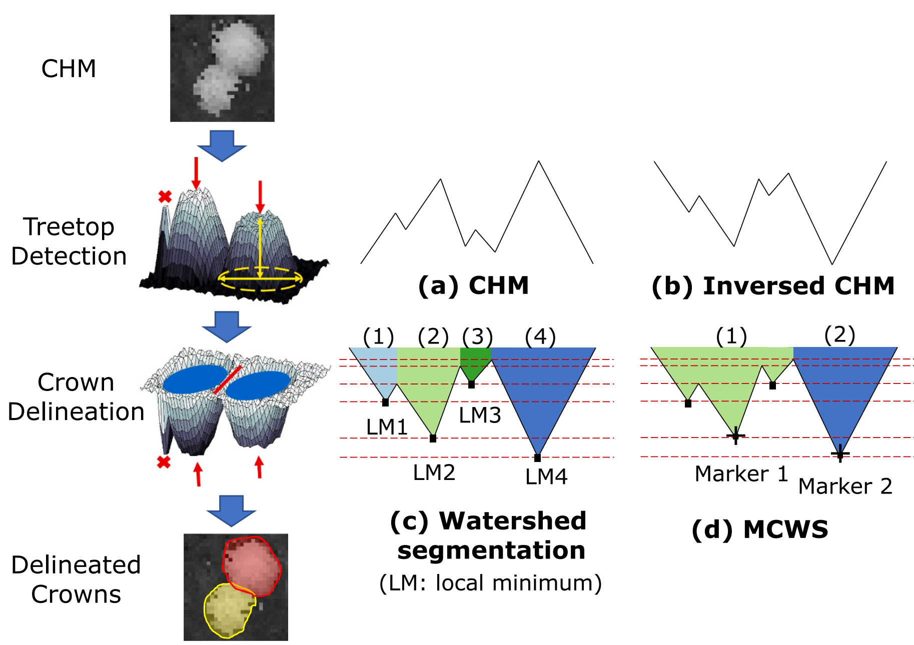

Fig. 4. Illustration of the MCWS algorithm for ITDD. In (c) and (d): each color represents a separate crown; the horizontal lines indicate where there is a local extremum (peak or valley). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

determine the potential crown size for each CHM pixel. Thirdly, we detected treetops by searching for the highest points in local windows, the size of which were decided according to the TH to CD relationship derived in the previous step. Finally, the crown boundaries were delineated using the MCWS algorithm with the detected treetops as the markers (Wang et al., 2004a; Yin and Wang, 2019).

Mask R-CNN (Fig. 5a): The Mask R-CNN method is a neural network with two main stages, i.e., region proposal and mask prediction. The region proposal stage is the backbone which generates feature maps and proposes regions that are likely to contain objects. The mask prediction stage is composed of two heads, including an object detection head that identifies the bounding box and class of each object and a mask generation head that provides segmentation results. In this study, we used the CHM as the input (replicated three times to make the required three-channel input) and exported the masks as the delineated crowns. To train this model, we cropped 12 subsets of size 32 m by 32 m from the

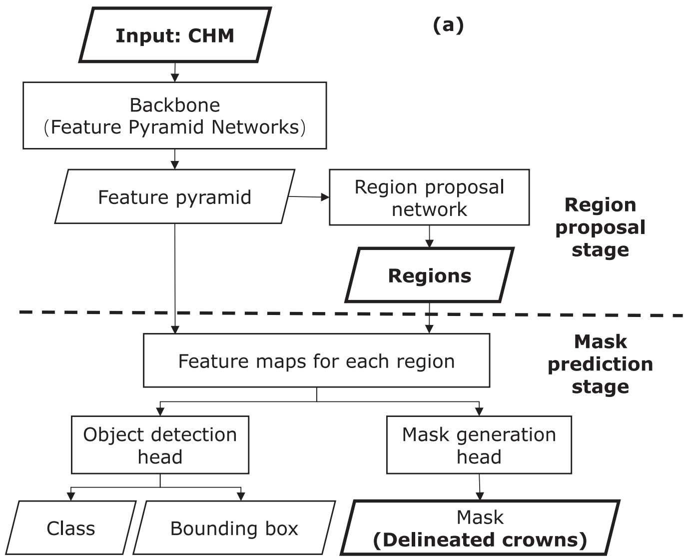

Fig. 5. Illustration of the deep learning algorithms for ITDD. (a) Mask R-CNN. (b) YOLOv7.

2019 CHM and manually labeled all the mangrove crowns in each subset. Ten plots were used for training and the other two were used for validation. The code by Pécot et al. (2022) was used to implement Mask R-CNN in this study (https://github.com/tpecot/NucleiSegmentationAndMarkerIDentification). We applied the parameters trained on the COCO dataset as the initial weights. The number of augmentations was set as 10 to enhance training data diversity. All default hyperparameters were maintained, except as specified. During the training process, we checked the model performance every 10 epochs, until no significant accuracy improvement was observed in the last 50 epochs. To apply the trained model, we first divided the entire CHM into 32-m-by-32-m subsets for input into the trained model. The segmentation results were regarded as the mangrove tree crowns and merged to create the ITDD results.

YOLOv7 (Fig. 5b): The YOLOv7 method is also a deep learning neural network. Unlike Mask R-CNN, YOLOv7 skips the region proposal step and directly goes to segmentation, which makes YOLOv7 more efficient. Instead of training a region proposal network, YOLOv7 divides the feature pyramid to smaller square tiles and then uses these tiles for object detection and mask generation. In this study, we used the CHM as the input (replicated three times to make the required three-channel input) and exported the masks as the delineated crowns. To train this model, we used the same training and validation data as for the Mask R-CNN model. The code by Wang et al. (2022) was used to implement YOLOv7 in this study (https://github.com/WongKinYiu/yolov7). To ensure a fair comparison between YOLOv7 and Mask R-CNN, the same initial weight parameters pretrained on the COCO dataset were also used for YOLOv7 training. The batch size was set to five, resulting in training data divided into two groups of five images each. For each epoch, the two groups were transformed and fed into the model one-by-one, which enriched the diversity of training data. The training process stopped after 500 epochs or if no significant improvement was observed over the last 100 epochs. Similar to Mask R-CNN, we divided the entire CHM into 32-m-by-32-m subsets and fed them to the model as input. The segmentation results were regarded as the mangrove tree crowns and merged to create the ITDD results.

(2) Individual tree matching

Different from field surveys, the trees identified through ITDD are not marked with the same ID across years. Therefore, the trees from different years must be matched before comparing their THs. The matching was according to the crown overlap rate between datasets. If two crowns from different datasets appear at the same location and exhibit an overlap exceeding a predetermined threshold, they are considered "matched", i.e., they correspond to the same tree crown (Yin and Wang, 2019). The illustrations of "matched" and not matched trees are provided in a later section 2.4.1.

To ensure the same trees are successfully matched over time, we first matched the ITDD crown segments with the crowns from the reference data to get the tree ID, and then link the ITDD crowns from different years by the tree IDs. The matched trees were then used for TH growth analysis.

(3) Individual TH measurement and growth estimation

The THs of individual trees were derived as the highest CHM value within each delineated crown.

Similar to the grid-based approach, the tree-based TH growth was both visualized and quantified. The overall statistical distributions of individual THs at the four data collection times were illustrated. Additionally, the average TH growth trajectory of mangroves from Nov 2015 to Jan 2019 was depicted. Via building a linear regression model with average TH as the dependent variable and year as the independent variable, we represented the average annual growth rate (in cm•yr^{-1}) by the slope.

2.2.3. Bias correction

It needs to be noted that the LiDAR-derived THs were not consistent over time because the LiDAR data for the four years were collected using different sensor types, of different point density, and pre-processed by different vendors. The base station for the RTK GPS collection also was changed from single station to network station. In addition, wind may have caused the TH to lower and vary during UAV data collection. These factors combined inevitably resulted in a bias in the LiDAR derived THs, which would introduce errors in the TH growth analysis. As summarized in Table 2, this potential bias has been vastly neglected in previous studies. Some studies attempted to minimize the bias by normalizing the DTMs derived from the multi-temporal LiDAR, referred to as the DTM normalization approach (Vepakomma et al., 2008). This approach, although still popular, aligns the datasets vertically without considering potential errors in heights above the terrain (Choi et al., 2019; Riofrío et al., 2022). Such errors have been recognized recently and was assumed to stem from differences in LiDAR point density. Based on a reasonable assumption that the THs get underestimated because treetops are missed in low-point-density LiDAR, the point density modeling

Table 2
Height correction approaches.

Approach	Assumption	References
Neglected	THs are correct	Yu et al., 2004; Srinivasan et al., 2014; Arumäe et al., 2020; McCarley et al., 2022
DTM normalization	THs are correct	Vepakomma et al., 2008; Choi et al., 2019; Riofrío et al., 2022
Point density modeling	Lower LiDAR point density causes underestimation because treetops are missed	Zhao et al., 2018

approach estimates TH by adding LiDAR point density as an explanatory variable. An exponential relationship between point density and TH has been found and used to correct the bias in TH (Zhao et al., 2018). Although this approach has proved effective in boreal forests, the error after correction (bias = 0.02 m, RMSE = 0.91 m (Zhao et al., 2018)) is still not negligible in the mangrove forests.

Therefore, we propose to use the height of a brick house roof (where the RTK base station was placed, Fig. 2d) as a benchmark for LiDAR height correction. Specifically, the elevation of the rooftop and the nearest ground were interpreted from the UAV-LiDAR data and compared to estimate the height of the brick house. Using the house height estimated in Jan 2019 as the baseline, the relative bias of the other three datasets were quantified. This relative bias was then subtracted from the respective CHMs to generate bias-corrected CHMs in Nov 2015, Nov 2016, and Nov 2017.

2.3. Mangrove TH growth affected by different factors

To quantify the TH growth rate of mangrove trees in different groups, we applied the multi-temporal change analysis method in section 2.2 separately to mangrove trees of three different species (Fig. 6a), in three clumpiness groups (Fig. 6b), and at two invasion statuses (Fig. 6c). Analysis of Variance (ANOVA) was conducted to examine the

Fig. 6. Typical appearance of different mangrove trees. (a1)-(a3) are the three dominant species. (b1)-(b3) are the three clumpiness groups. (c1)-(c2) are mangrove trees invaded by *Spartina alterniflora*.

correlation between mangrove TH growth rates and the three factors independently.

2.3.1. Species

For the analysis of TH growth rates across different species, we referenced the species information for 236 trees, which remained consistent over time. Using the method outlined in section 2.2, we analyzed the LiDAR derived THs for each species individually, i.e., *Avicennia marina* (112 trees), *Aegiceras corniculatum* (68 trees), and *Kandelia obovata* (40 trees). In addition, to assess whether the growth rates differed significantly among these species, we summarized the per-tree growth rate for each species and conducted ANOVA to test the statistical significance.

2.3.2. Clumpiness

We separated the reference mangroves into three clumpiness groups. Crowns not connected to any other crowns are in the “isolated” group; crowns partially connected to other crowns and partially exposed to the empty space from nadir view are in the “boundary” group; crowns connected to other crowns in all directions are in the “clumped” group. The method in section 2.3.1 was then applied to the LiDAR derived THs in each clumpiness group to calculate the different TH growth rates. Among the tree samples, we had 117 isolated trees, 232 boundary trees, and 73 clumped trees respectively.

2.3.3. Invasion

The number of mangrove trees affected by invasion changed over time. In order to analyze the impact of invasion on mangrove TH growth rate, we consider mangrove trees that have always been surrounded by or close to *Spartina alterniflora* as “invaded” and those never near *Spartina alterniflora* as “not-invaded”. The method in section 2.3.1 was then applied to LiDAR derived THs in these two groups to calculate the different TH growth rates. Among the tree samples, we had 121 trees that were invaded by *Spartina alterniflora* and 253 trees at least 5 m away from the *Spartina alterniflora*.

2.4. Accuracy assessment

2.4.1. Individual tree detection accuracy assessment

We used the widely employed summary metric, detection accuracy (DA) (Yin and Wang, 2016), to assess the accuracy of the individual mangrove tree detection. DA is determined as the ratio of the number of correctly delineated crowns to the total number of reference crowns, i.e.,

$$DA = \frac{N_{\text{correct}}}{N_{\text{total}}} \tag{1}$$

where N_{total} is the number of reference crowns and N_{correct} is the number of correctly delineated trees. A crown is considered “correctly delineated” if it overlaps with a reference crown by over a certain threshold, as illustrated in Fig. 7. This threshold is often set as 50% to identify perfect matches (Leckie et al., 2003; Yin and Wang, 2019), while lower matches have also been accepted (Yin and Wang, 2016; Niccolai et al., 2010). In this study, we set the threshold to a more lenient value 30% to accommodate possible spatial displacements.

2.4.2. TH measurement accuracy assessment

Accurate TH measurement is fundamental for TH growth monitoring. In this sudy, we checked each correctly delineated individual tree that has field measured TH data. For each year, a scatterplot showing the correspondence of the THs based on UAV-LiDAR and field campaigns was created to provide an overall assessment of the TH measurement. Two standard metrics, bias and root mean square error (RMSE), were adopted to quantify the accuracy (Yin and Wang, 2016).

$$bias = \frac{1}{n}\sum TH_{est} - TH_{obs} \tag{2}$$

$$RMSE = \sqrt{\frac{1}{n}\sum (TH_{est} - TH_{obs})^2} \tag{3}$$

Here, n is the number of available samples. TH_{est} and TH_{obs} are the THs based on UAV-LiDAR and field campaigns respectively.

2.4.3. TH growth rate accuracy assessment

The accuracy of TH growth rate estimation was also directly assessed by comparing the growth rates obtained from field surveys and from UAV-LiDAR. Given that the UAV-LiDAR data cover a much larger number of trees than field measurements, a direct comparison of the average rates might be misleading. Hence, our assessment focused on the growth rates of THs that were both field-measured and LiDAR-estimated. The accuracy metrics used were bias, RMSE, and Pearson’s correlation coefficient. The statistical significance of correlation was also checked.

3. Results

3.1. TH measurements

3.1.1. ITDD results

Noting that the setting of overlap rate thresholds for tree matching would change the ITDD and TH measurement accuracy, we tried different thresholds (from 0 to 100%, every 1% steps). As expected, the DA decreased with increasing thresholds (Fig. 8). Regardless of the threshold, the MCWS and Mask R-CNN methods always had similar performance, while YOLOv7 consistently had higher DA, making YOLOv7 the optimal model for mangrove ITDD. With 30% as the threshold, the individual mangrove trees were identified with with 78.5% DA on the Nov 2016 data and 87.2%–90.4% DA in the other three years. These crowns were used in the subsequent tree-based approach analyses.

Among the correctly delineated trees, 58 trees in Nov 2015, 79 in Nov 2016, 175 in Nov 2017, and 173 in Jan 2019 had field measured THs. Using the UAV-LiDAR data, the THs were overestimated by 11 cm in Nov 2015, and underestimated by 4 cm, 9 cm, and 8 cm in Nov 2016, Nov 2017, and Jan 2019, respectively (Fig. 9). The RMSEs were in the range of 20–27 cm.

3.1.2. Bias correction

The LiDAR profiles near the brick house (Fig. 10) revealed varying roof heights in the four years’ datasets: 3.00 m, 2.75 m, 2.63 m, and 2.63 m, respectively. Using the data in Jan 2019 as the baseline, we found

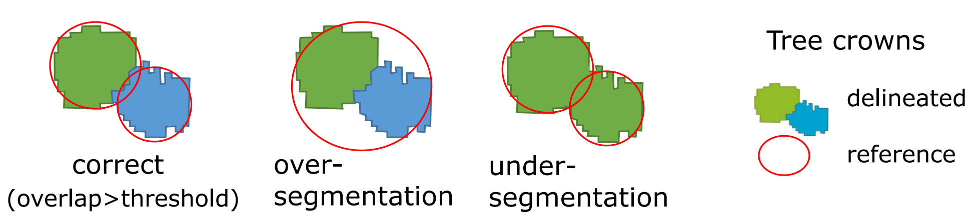

Fig. 7. Criteria for determining the correctness of ITDD. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

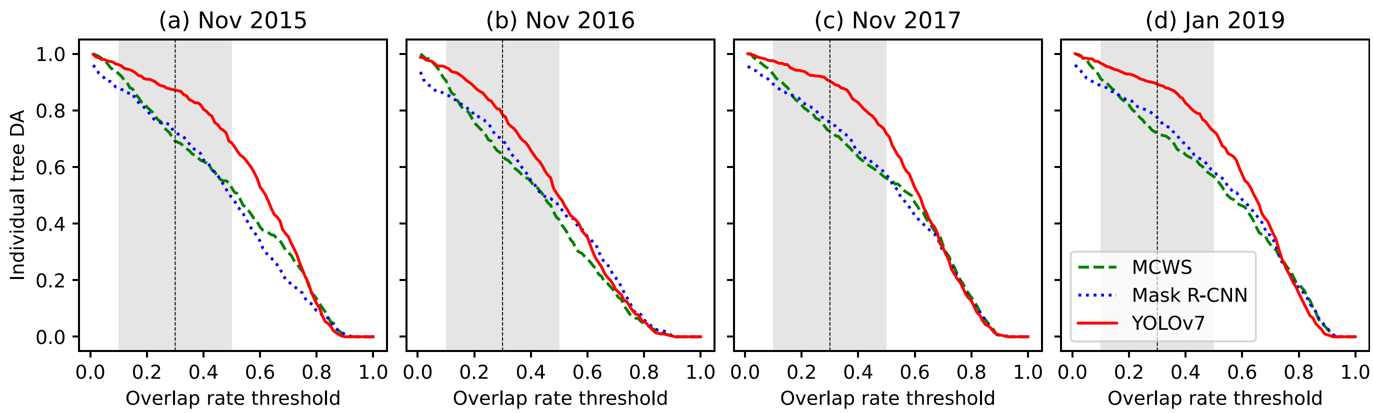

Fig. 8. ITDD accuracies of the three algorithms in (a) Nov 2015, (b) Nov 2016, (c) Nov 2017, and (d) Jan 2019. The gray shadow area highlights overlap rate threshold between 10% and 50%, while the dashed vertical line marks 30%.

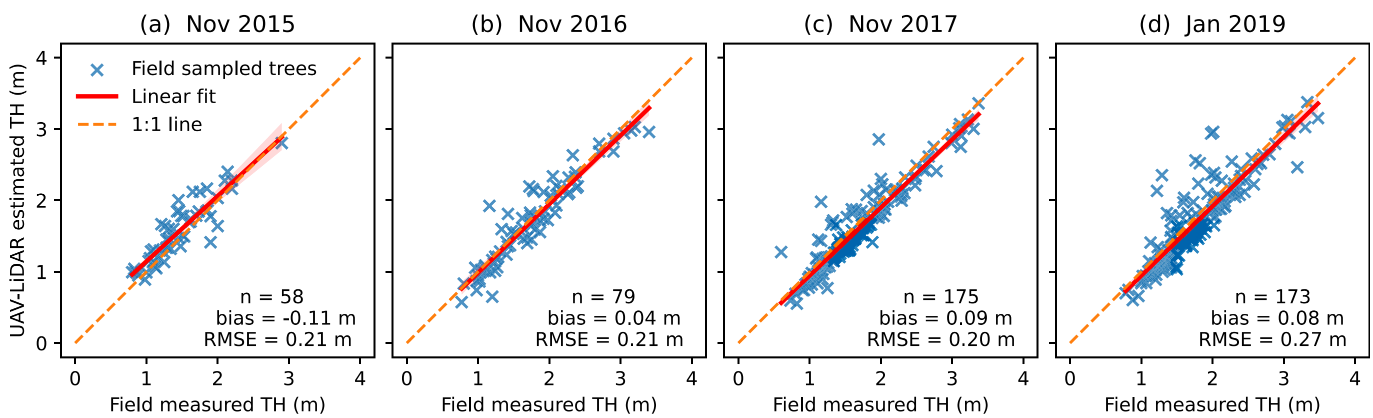

Fig. 9. Individual TH measurement accuracy using YOLOv7 (tree matching at 30% overlap rate threshold) in (a) Nov 2015, (b) Nov 2016, (c) Nov 2017, and (d) Jan 2019.

that the LiDAR-estimated THs in Nov 2015, 2016, and 2017 should be adjusted by −0.37 m, −0.12 m, and 0.00 m, respectively. This general trend that THs in Nov 2015 were overestimated the most and the data in Nov 2017 agreed well with that in Jan 2019 is consistent with the TH assessment findings in section 3.1.1, which implies the validity of the bias correction step.

3.2. Growth rate estimation accuracy

The growth rate estimation accuracy was summarized in Fig. 11 and Table 3. Depending on the years of data used, the number of tree samples and the accuracy were different. In Fig. 11(a), the blue color represents data from trees measured and correctly delineated in both Nov 2017 and Jan 2019, totaling 156 trees. When we extended this comparison to 2016 (green) and 2015 (brown), the number of applicable trees dropped to 67 and 13, due to limited field measurements in 2015 and 2016 caused by logistical challenges.

The estimated and field measured growth rates were compared at both the individual tree level and the summarized level. At the individual tree level (Fig. 11a), most of the estimations were close to the field measurements, with a few outliers. Despite the high variation, a significant positive correlation between the UAV-LiDAR estimated growth rates and field measurements was observed (Table 3). At the summarized level (Fig. 11b), the bias was minimal at 0.7 cm•yr^{-1} when only the Nov 2017 and Jan 2019 data were used, which domenstrated the validity of the method. Extending the comparison to Nov 2016 and Nov 2015, the bias increased to 4.1 and 4.0 cm•yr^{-1}, respectively, which was rather high considering the low growth rate. This higher bias compared to the Nov 2017 to Jan 2019 scenario could probably be attributed to errors in the bias correction step. Another contributing factor to the accuracy disparities was likely the distinctions among LiDAR sensors (Table 1). In addition, it should be noted that different trees were included in the accuracy assessment, which could also have added to the varied performance. Interestingly, as the analysis period extended, extreme samples got excluded and the RMSE actually decreased. Nevertheless, the overall distribution were similar between the field measurements and the UAV-LiDAR estimations (Fig. 11b), which implied that the qualitative analyses and subsequent comparisons among different mangrove groups are still feasible and valid.

3.3. Multi-temporal change analysis

3.3.1. Growth rate from the grid-based approach

Fig. 12 illustrates the UAV-LiDAR-derived canopy heights in the entire study area and their growths. Each gray line shows the growth trajectory of one CHM pixel. These lines generally indicated an increasing trend in canopy heights although some pixels had decreased heights. Possible reasons of the decreases include actual height decreases and errors caused by spatial displacement. The red squares and line denote the average TH and its change over time. The average TH increased from 1.78 m to 2.31 m, leading to an estimated average TH growth rate of 17.5 cm•yr^{-1}.

3.3.2. Growth rate from the tree-based approach

Fig. 13 shows the results of TH growth analysis using the tree-based approach. If we check each individual tree, the change of TH between

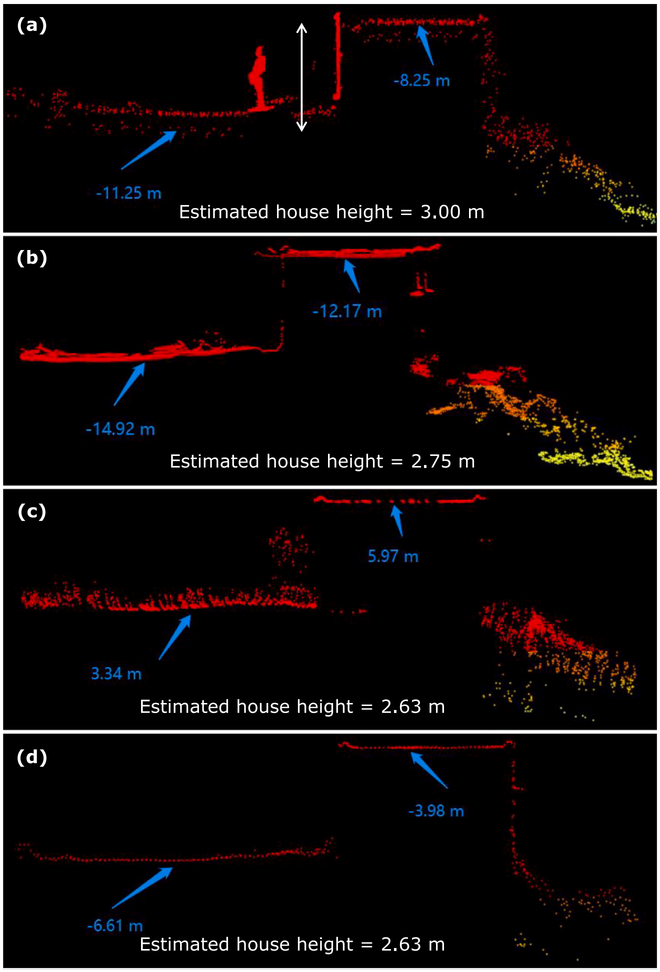

Fig. 10. Bias estimation according to the brick house roof height. (a)-(d) are the LiDAR profiles of the house in Nov 2015, Nov 2016, Nov 2017, and Jan 2019.

consecutive years varied from −55.8 cm to +139.3 cm (Fig. 13a), with 95% of these changes falling in the −28.7 to +58.5 cm interval. Yet the average change of TH had a much smaller magnitude, probably due to the even out between taller and shorter trees. If we assume a constant growth rate, the annual growth rate of individual mangrove trees in the study site ranged from −10.5 cm•yr^{-1} to 42.1 cm•yr^{-1} (Fig. 13b).

The distribution of UAV-LiDAR-derived THs is in Fig. 13c. The gray area shows the histogram of TH for each year, while the four dots and the connecting solid line show the average TH and its growth trajectory over time, respectively. Through matching the UAV-LiDAR delineated crowns with the field samples, the correspondence for 281 trees were confirmed. The average growth rate of these trees was estimated to be 12.1 cm•yr^{-1}.

3.4. Factors affecting mangrove TH growth

3.4.1. Species

Different growth rates were discerned among the three dominant species (Fig. 14). The fastest growing species was *Avicennia marina*, at 15.5 cm•yr^{-1} (52 trees). The average TH growth rate of *Aegiceras corniculatum* (30 trees) was 11.7 cm•yr^{-1}, while that of the *Kandelia obovata* (26 trees) was 10.7 cm•yr^{-1}. The ANOVA result indicated that these differences among different species was statistically significant ($p < 0.05$). Checking the distribution of the growth rates (Fig. 14b), it was found that not all trees had positive growth. Although the majority grew taller, still a small number of trees decreased in height.

3.4.2. Clumpiness

The analysis regarding the effect of clumpiness on mangrove growth rates suggested that trees in denser clumps grew faster (Fig. 15). The average growth rate of the isolated (79 trees), boundary (147 trees), and clumped (36 trees) were 9.3 cm•yr^{-1}, 12.6 cm•yr^{-1}, and 18.3 cm•yr^{-1}, respectively. The ANOVA result indicated a statistically significant difference in growth rates among different clumpiness groups ($p < 0.001$). Checking the growth rate distributions (Fig. 15b), it was noted that trees in both the isolated and boundary groups had some trees with reduced heights while all trees in the clumped group got taller.

3.4.3. Invasion

Trees of different invasion statuses also had different TH growth rates (Fig. 16). The average growth rate of the 97 *spartina*-invaded trees was 13.1 cm•yr^{-1}, while that of the 133 not-invaded ones was 12.0 cm•yr^{-1}. This suggests a possible association between invasion and higher growth rates. However, the ANOVA result indicated that this difference was not statistically significant ($p = 0.358$). Checking the growth rate distribution, we found that both invaded and not-invaded groups had some trees that got shorter (Fig. 16b).

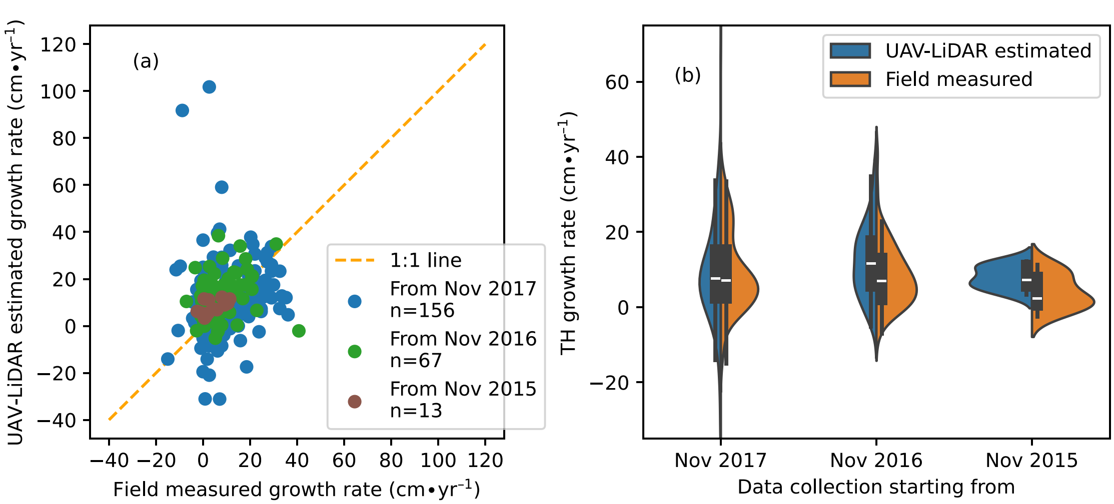

Fig. 11. Comparison between UAV-LiDAR estimated and field measured growth rates. (a) Individual tree growth rates. (d) Overall distribution. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Summary of growth rate accuracy.

Data used	Number of tree samples	Growth rate (cm•yr^{-1})				Pearson's correlation	*p*-value
		UAV-LiDAR estimated	Field measured	bias	RMSE		
Nov 2017, Jan 2019	156	10.1	9.4	0.7	17.8	0.17	0.03
Nov 2016, Nov 2017, Jan 2019	67	12.4	8.3	4.1	11.6	0.28	0.02
Nov 2015, Nov 2016, Nov 2017, Jan 2019	13	8.0	4.0	4.0	5.3	0.57	0.04

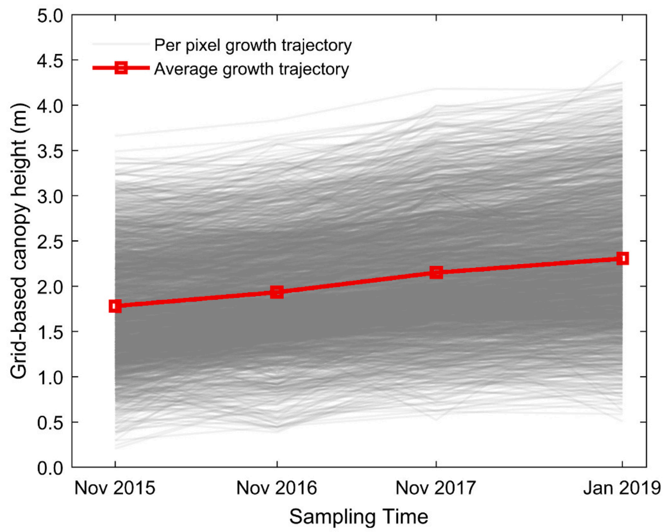

Fig. 12. Canopy height at the different data collection time from the grid-based approach.

4. Discussion

4.1. Significance

This work tackled the challenge of accurately estimating individual mangrove TH growth by exploiting multi-temporal UAV-LiDAR, making a pioneering effort in monitoring forest TH growth with multi-temporal UAV-LiDAR. Although multi-temporal airborne LiDAR has been used in some forest types, UAV-LiDAR has not been investigated. The rationale for choosing UAV-LiDAR over airborne ones is its superior point density and accuracy, which are crucial for mangrove studies where growth rates are much lower (9.3–18.3 cm•yr^{-1}) (this study) than those of other terrestrial forest types (> 20 cm•yr^{-1} growth rate) (Yu et al., 2008; Zhao et al., 2018). To this end, we derived two key findings: (1) The tree based approach, which closely resembles field survey methods compared to the grid-based approach, is effectively applicable for TH growth monitoring using multi-temporal UAV-LiDAR. (2) Statistically significant differences in growth rate has been found among different mangrove species and clumpiness groups, but not evident between different invasion statuses. In the following sections, we will delve into detailed discussions about these findings and limitations.

4.2. Findings

The optimal method for mangrove TH growth analysis involves collecting UAV-LiDAR data, individual TH measurement, and multi-year TH comparison, as illustrated by the solid boxes and arrows in Fig. 17. For optimal data collection, UAV-LiDAR data should be obtained from the same sensor (e.g., Riegl VUX-1LR). For data processing, we recommend adopting the tree-based approach to derive the THs of individual trees from each dataset, which can then be compared to obtain the TH growth rate. If the overall growth rate in the study area is needed, a direct comparison of the average THs is enough. In addition, tree matching between the multi-year individual tree crowns can be added to obtain more detailed TH growths (black dashed lines and boxes in Fig. 17).

In case the UAVs, sensors, or RTK-GPS settings are different during

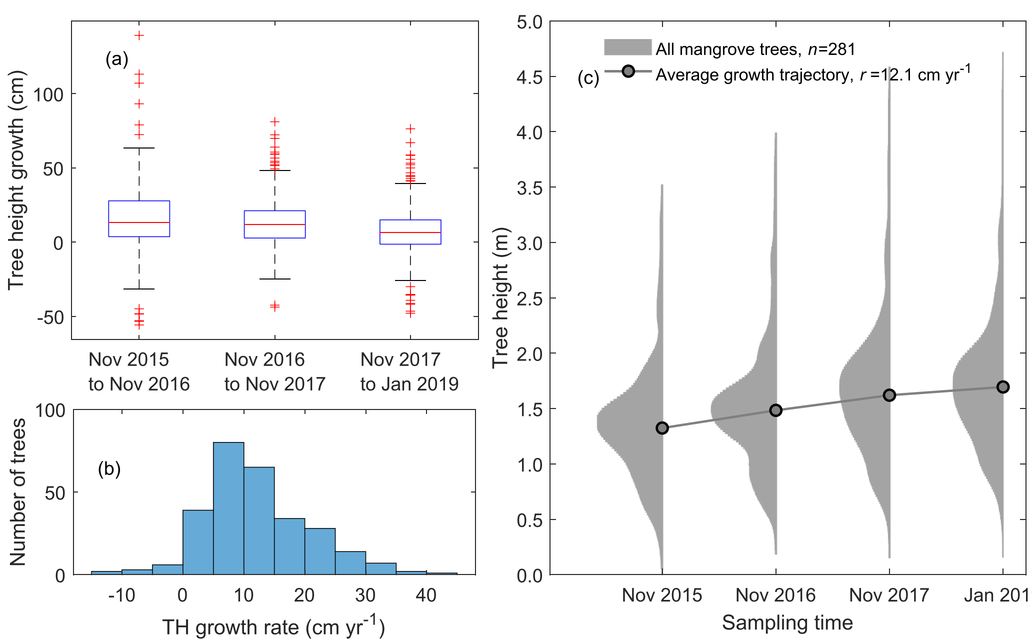

Fig. 13. Estimated mangrove growth from the tree-based approach. (a) Boxplot of individual TH growth. (b) Histogram of individual TH growth rate. (c) Violin plot of the THs.

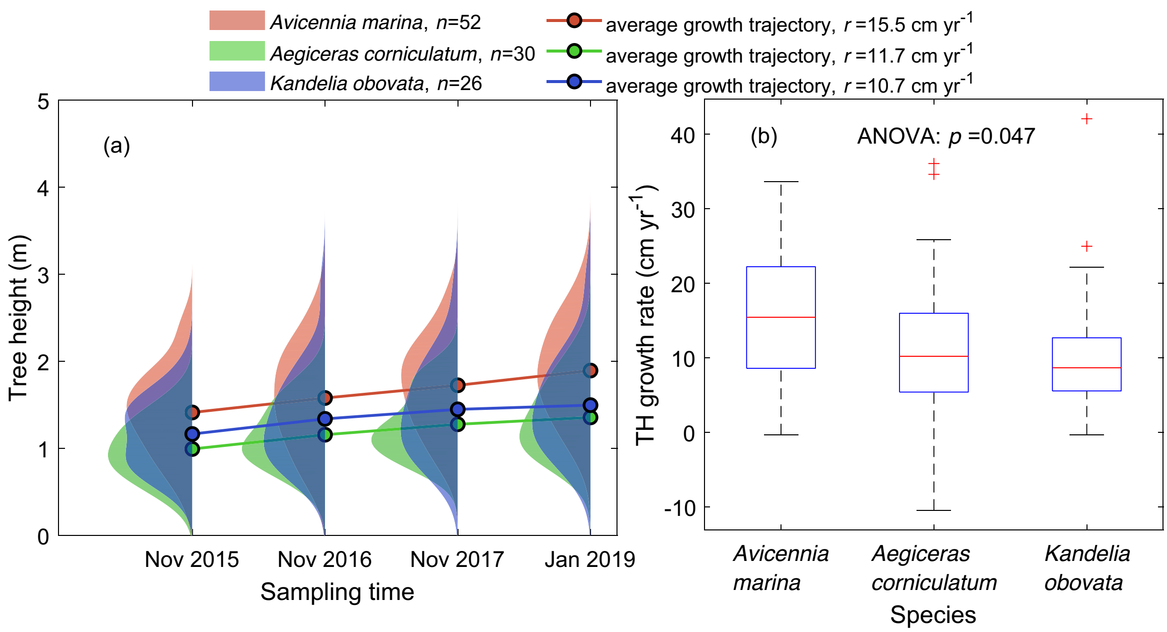

Fig. 14. Growth of different mangrove species, based on multi-temporal LiDAR analysis. (a) Violin plot of the THs. (b) Boxplot of the individual TH growth rates of the three species. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

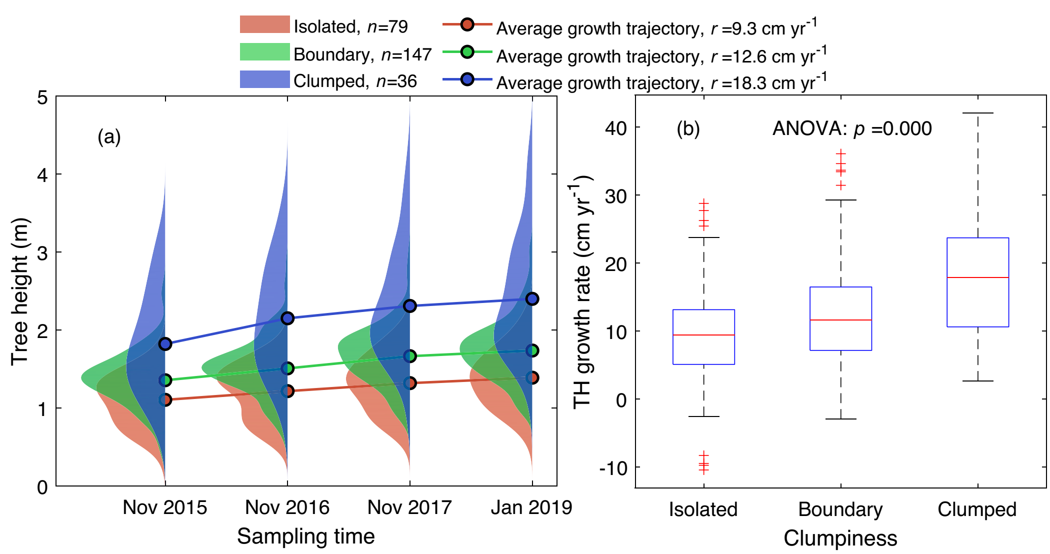

Fig. 15. Growth of mangroves in different clumpiness groups, based on multi-temporal LiDAR analysis. (a) Violin plot of the THs. (b) Boxplot of the individual TH growth rates of each clumpiness group. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

data collection, we propose two additional optional steps to further improve the accuracy: co-registration and bias correction (gray dashed arrows and boxes in Fig. 17). The co-registration step has been vastly ignored in previous LiDAR-based forest change analysis, probably because the spatial displacement is marginal compared to the tree size and LiDAR point density. However, as UAV-LiDAR point density gets increasingly higher (easily >1000 pt./m^2 now), LiDAR-based analysis is quickly moving towards the ultra-high spatial resolution scenario, where accurate co-registration is indispensable. Particularly as the mangrove trees in the study are of small crown width (the smallest is only 1 m), spatial displacement becomes a non-negligible factor. The bias correction step, although less discussed in the literature, is equally important. Height measurement errors used to be large (e.g., 1.5 m in Zhao et al., 2018). With different sensors, this bias can vary over time, which will cause significant errors in TH growth analysis. Therefore, bias correction based on invariant ground features within the study area is useful to validate the LiDAR-based heights.

4.2.1. UAV-LiDAR data is accurate for mangrove TH growth measurement

UAV-LiDAR data proves accurate for mangrove TH growth measurement, under the premise that the same high-accuracy LiDAR sensor is used for multi-temporal data collection. In our study, the deviation of THs estimated from UAV-LiDAR data from the corresponding field measurements varied between −9 cm and +11 cm (Fig. 9). Compared to the slow TH growth rate (12.1 cm•yr^{-1}) of the mangrove trees in the area, this 20 cm variation implied that TH growth estimates within four years could be erroneous without proper bias correction. Despite so, this pitfall can be bypassed by using the same UAV-LiDAR equipment type for multi-year data collection. In our experiment, the deviations between LiDAR and field data were −9 cm and −8 cm respectively for 2017 and 2019 when data were collected using the same sensor, Riegl VUX-1LR. This resulted in a minimal systematic error of just 1 cm when

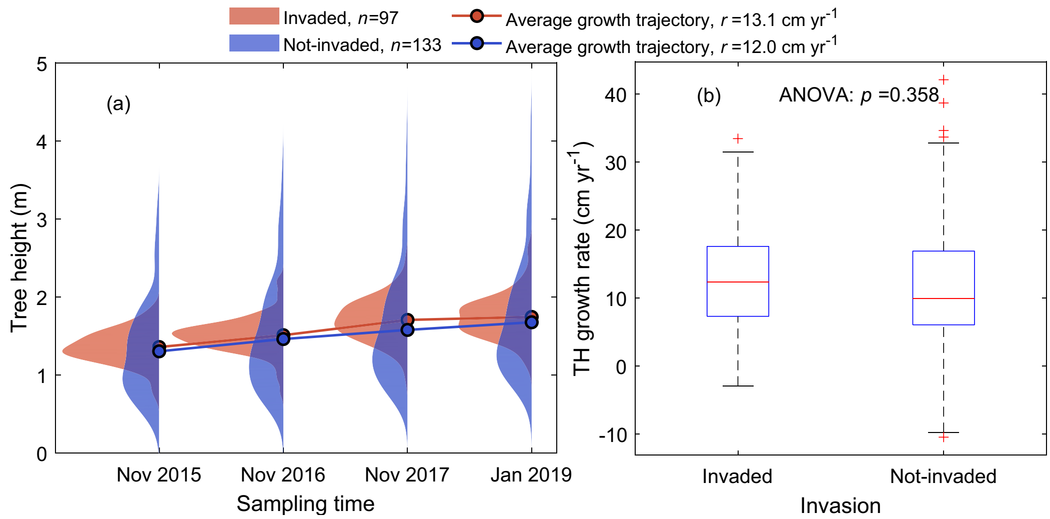

Fig. 16. Growth of mangroves in different invasion groups, based on multi-temporal LiDAR analysis. (a) Violin plot of the THs. (b) Boxplot of the individual TH growth rates of the two invasion groups. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

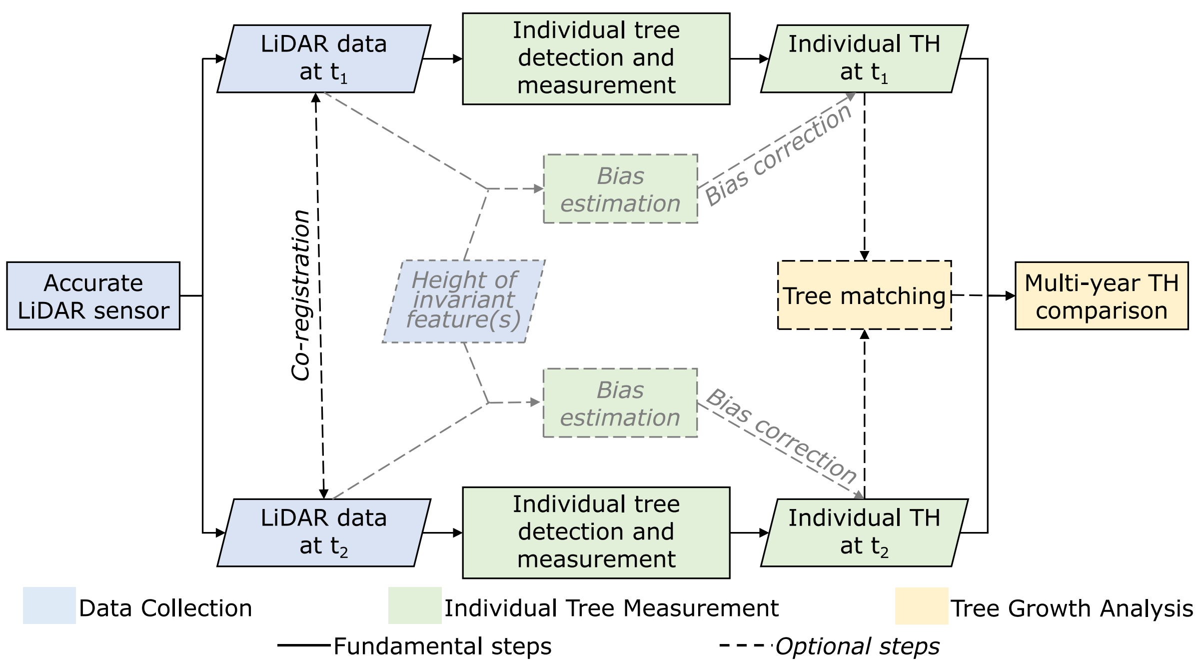

Fig. 17. Recommended diagram for UAV-LiDAR-based TH growth rate measurements. Dashed boxes, dashed arrows, and italic fonts represent optional steps in the procedure. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

comparing the two datasets for TH growth analysis, thereby accurately representing the annual TH growth. It is important to note the role of the inertial measurement unit (IMU) in ensuring the data accuracy, necessitating the use of an IMU of the same or higher quality in the operations.

Previous studies have demonstrated the feasibility of using multi-temporal LiDAR to measure individual tree growth over half a year or longer time intervals (Zhao et al., 2018). Tree growth of 0.24 m to 0.8 m have been reported measurable by multi-temporal airborne LiDAR (Hopkinson et al., 2008; Hudak et al., 2012; Zhao et al., 2018). However, because the mangrove growth rate in our study area is as low as 0.12 m/yr, these airborne LiDAR datasets can only detect mangrove growth at intervals of over two years. Compared to these studies, the 1 cm relative deviation between the two UAV-LiDAR datasets in 2017 and 2019 in our current study indicates that UAV-LiDAR data have greater potential than airborne LiDAR for monitoring mangrove TH growth, as well as in other forests with slow growth rates.

4.2.2. *Tree-based approach is a feasible substitute for field survey*

The tree-based approach proposed in this study for mangrove TH growth monitoring has demonstrated its effectiveness through accurate tree detection and height measurement. The success of this work highlights the possibility of boosting field surveys for mangrove ecology research and mangrove forest management with remote sensing. Traditionally, field-observed growth rates based on yearly measurements of 60 trees requires several days of work by two to three people each year. In contrast, UAV-LiDAR data for the entire study area could be collected in half an hour, which greatly reduces the need of physical entrance into the mangrove woods while significantly improving the monitoring efficiency.

The tree-based approach and the grid-based approach led to different growth rate estimates. Although literature directly comparing tree-based and grid-based TH growth estimations is rare, similar findings have been implied. For example, Zhao et al. (2018) found that forest loss was well represented in the grid-based approach, which we could anticipate to largely degrade the estimation of TH growth rate. The disagreement between the tree-based and grid-based growth rate estimates could be partly explained by their differing definition of "height". In the tree-based approach, the height being evaluated is the height of treetops, which corresponds well to the natural definition of TH. It should be noted that the tree matching criteria in this approach can vary, which might lead to slightly different estimation results. In the grid-based approach, the height of all pixels in the CHM are considered, which include not only the actual TH, but also the height of other branches. Moreover, changes of height in the non-mangrove or non-vegetated pixels could also be included, which makes the grid-based results further deviated from the desired tree dynamics information.

4.2.3. Factors affecting mangrove TH growth

Mangrove tree growth rate is significantly affected by many environmental factors. In this exploratory work to investigate mangrove TH growth with remote sensing, we checked three factors that may witness different growth rates, i.e., species, clumpiness, and invasion. Both significant (species and clumpiness) and non-significant (invasion) variations were found.

Tree growth rates vary by species and sites, as is supported by the findings of this study as well as the literature. To our best knowledge, growth rates in the Shankou Mangrove Reserve has been rarely reported in the literature. Papers reporting the growth rate of mangrove trees near our study site were not found. In the Maowei Sea area, which is also part of the Beibu Gulf and about 100 km to the west of our study site, the *Aegiceras corniculatum* community of around 2 m TH had over 10 cm annual TH increment (Du and Zhang, 2020). Particularly in the landward fringe which is more similar to our study, the annual growth rate was 10 ± 4 cm for the 14 year old community and 13 ± 5 cm for the 8 year old community. The growth rate of *Aegiceras corniculatum* in our study was found to be 11.7 cm•yr^{-1}, which is of comparable magnitude. In addition, a field-survey-based study in the Futian Mangrove Reserve of Shenzhen in Guangdong, China (about 600 km to the east of our study site) reported average growth rates of 18.2 cm•yr^{-1}, −2.2 cm•yr^{-1}, and 21.7 cm•yr^{-1} and maximum growth rates of 90 cm•yr^{-1}, 75 cm•yr^{-1} and 95 cm•yr^{-1} for *Avicennia marina*, *Aegiceras corniculatum*, and *Kandelia obovata,* respectively, from 1994 to 1996 (Liao et al., 2004). The average growth rates detected in our study for the three species are much lower (15.5 cm•yr^{-1}, 11.7 cm•yr^{-1}, and 10.7 cm•yr^{-1}), possibly be due to differences in environmental factors and mangrove growth stages. Our results thus supplement the growth rate database for these three species, especially for the Guangxi Shankou National Mangrove Nature Reserve.

The observed increase in growth rate with clumpiness was unexpected as isolated trees are often presumed to face less competition from their fellows. According to our limited field surveys, we infer that one possible reason why the isolated trees grew slower is that they have less protection from tides and strong winds. This is consistent with a finding that understory mangrove trees below clumped upperstory trees are better protected and lost less biomass after typhoons (Chen, 2021). Among ecological studies, some claimed that trees growing with larger gaps would have higher growth rate (Imai et al., 2009). However, that assumption may not apply to the mangrove forest because they have different ecological traits compared to other forest types. A recent work found that the relative growth rate declined as mangrove cover increased (Peng et al., 2022), which contradicts our findings. Nevertheless, their study was in a different continent on a different mangrove species (*Avicennia germinans*) and by manually thinning the plots to change the mangrove cover. In other words, the conflicts in our findings could be caused by various factors. We believe that more studies like ours should be conducted to further investigate the association between clumpiness and mangrove growth rate.

No significant difference of growth rates was found between the mangrove trees invaded by *Spartina alterniflora* and those not invaded. Research on the relationship between mangrove TH and invasion was rare. Most studies on species invasion in the mangrove forests are descriptive of what and where the invasion happened (Biswas et al., 2018; Lu et al., 2022). Both positive and negative impacts of *Spartina alterniflora* on mangroves have been found (Chen and Ma, 2015). The good impacts include increasing the biodiversity and improving the resistance to heavy metal pollution, etc. The negative impacts include decreasing soil organic carbon, altering soil bacterial communities, etc. We did find one article concluding that *Spartina alterniflora* was beneficial for the growth and survival of seedlings in a downstream euhaline site in the Zhangjiang Estuary area (Zhang et al., 2012). Nevertheless, the mechanism is not clear. In addition, factors such as invasion may need longer time to affect the growth rate of mangrove trees, which we suppose partly explains why the difference between invaded and non-invaded trees were not statistically significant in our study. Examinations that include more tree samples, preferably over a longer time period and at larger spatial scales, would be interesting to further analyze the impact of invasion on mangrove TH growth.

4.3. Limitations and potential solutions

Since both the data and the topic in this work are new, the methods and findings can be insufficient. This section discusses the limitations of the study in three aspects: data source, methodology, and results. The possible causes and solutions are also analyzed.

4.3.1. Data source

THs derived from different LiDAR datasets are not always comparable over time. In this study, the four-year LiDAR data were collected using three different LiDAR sensors. When the research was first conducted (in 2015), UAV-LiDAR was not a quite affordable data source. Hence, we relied on external companies for data collection. We tried to use the best available devices to our knowledge in each survey. As a result, different LiDAR scanners were included and different flight designs were adopted. The shift from a single base station to network base stations for the RTK-GPS measurements could have introduced additional uncertainties in the multi-temporal analysis. More investigations should be conducted to better understand the impacts of these factors. Nevertheless, we would like this work to provide a way for optimizing the use of multitemporal LiDAR data that were collected under non-optimal conditions.

As a result of the above settings, the data provided to us contained different error in height. For example, the THs from LiDAR were respectively 9 cm and 8 cm shorter than the field measurements in 2017 and 2019, but 11 cm taller in 2015. Therefore, adjustment must be done when different data were used. Moreover, because the work is at such high spatial resolution, a tiny displacement could lead to big difference. Hence, the agreement between different datasets should be checked even if the same sensor and pre-processing were applied. For researchers who may want to adopt the approach proposed in this study, we recommend setting up non-change targets as reference for data height bias adjustment.

4.3.2. Methodology

Registration between different UAV-LiDAR datasets is critical for their multitemporal anlaysis. Although the nominal ranging accuracy is 2 cm, the horizontal displacement between LiDAR point clouds can be much larger (e.g., 3.05 m in this study). Existing registration methods are often designed for engineering scenarios where flat surfaces, regular shapes with sharp angles, and invariant features are dominant (Cheng et al., 2018; Huang et al., 2021). In contrast, the registration of different LiDAR datasets in forest change studies have been largely omitted, possibly because the spatial displacement is negligible compared to the spatial scale of the study (e.g., a large tree with over 20 m crown width or an entire forest stand) (Arumäe et al., 2020; Leipe and Carey, 2021; Vepakomma et al., 2008). However, stepping into the fine scale of individual trees, even minor spatial displacements between datasets can result in incorrect tree matching, leading to inaccurate TH growth estimations. Therefore, to accurately register LiDAR datasets from different time or sensors in the densely forested area is worth to work on and will benefit the study of multi-temporal LiDAR analysis. This can be particularly challenging due to changes in tree structure or environmental conditions over time. To mitigate these issues, future work may focus on developing more robust tree matching techniques that can accurately align trees across different datasets. This could involve advanced spatial analysis algorithms or machine learning models trained on diverse forest datasets.

The correlation between the measured and estimated growth rate, although statistically significant ($p < 0.05$), was not very strong. Besides ensuring sensor consistency, this issue may benefit from the following aspects. First is to implement more sophisticated preprocessing techniques, such as advanced noise filtering and point cloud normalization, so as to enhance the quality of the LiDAR data. Second is to develop more accurate algorithms for tree matching across different datasets. This could involve machine learning techniques that learn from a variety of tree shapes and sizes. Third is to integrate with other remote sensing data. Combining UAV-LiDAR data with other forms of remote sensing, such as multi-spectral or hyperspectral imagery, may provide additional insights into tree health and growth patterns, thereby improving growth rate estimates. Additionally, extending the duration of study to include more years of data can provide a more comprehensive understanding of growth patterns, potentially leading to more accurate growth rate estimations.

One other limitation in the current study is that the species-specific TH growth was limited to field measured trees because we did not identify species from the LiDAR datasets. To generalize the approach to large-scale applications, automated identification of tree species from remote sensing is necessary. We anticipate two possible solutions to this problem: (1) to classify species based on the differences in the 3D structure of the tree species, and (2) to combine LiDAR with multi-spectral reflectance images for more accurate species classification.

4.3.3. Results

The results should be interpreted with the awareness that what is observed is the change in TH rather than the growth of TH. Some trees got shorter, which contradicts with our intuition that “trees grow taller”, mainly because the detected “growth” is the net change of TH instead of pure growth. During our field campaigns, we noticed that some tree branches got broken by natural factors (such as tide, wind, birds, and worms) which caused some trees to get shorter. Literature also states that the mangrove trees experience a “self-thinning” process where the tree branches naturally break off (Liao et al., 2004). This decrease of TH counteracted the growth, and resulted in a net height change that is usually lower than the actual growth, and sometimes negative. Such complexities, which cannot be avoided in either remote-sensing-based TH estimation or field TH measurement, should be acknowledged and accounted for in anlaysis and interpretation.

4.4. Further prospects

Besides tackling the abovementioned limitations, this study can still be improved further. Here we propose prospects in two different pathways.

First, to analyze the influence of more factors on mangrove TH growth. Under natural conditions, the salinity, inundation, and temperature are some widely reported factors that affect the mangrove growth. In addition, due to the intertidal location of the mangrove forests, extreme weather events such as hurricanes change the mangrove height abruptly. Moreover, anthropogenic factors such as pollution runoff, hydrology alteration, and predation of propagules are also threatening the health of mangroves. Although policies are being made to protect the mangrove ecosystems, whether and why certain actions are good or bad to the mangrove growth and reproduction are still debated broadly. Among these factors, some (e.g., distance to sea water, flooding frequency, and initial TH) are easy to estimate using remote sensing. To use the LiDAR estimated TH growth as the dependent variable and analyze its relationship with some more localized factors (e.g., nutrient level) is also a promising follow-up work. In addition, due to limited access to different information (e.g., nutrient level, salinity) and the relatively small sample size, we could not check the intercorrelation of multiple environmental factors or their combined effect on mangrove TH growth. It would be interesting to build regression models with interaction terms in the future to comprehensively analyze the response of mangrove TH growth to various environmental factors.

Second, to automatically quantify the change of substrate elevation in order to improve the TH growth estimated from the LiDAR-derived CHMs. The average vertical accretion rate of the substrate in the mangrove area has been reported to be between −0.02 and 0.34 $cm{\bullet}yr^{-1}$ (Bird and Barson, 1977; Furukawa et al., 1997; Krauss et al., 2003; Li et al., 2015), and can be as high as 20 $cm{\bullet}yr^{-1}$ under extreme circumstances (Sidik et al., 2016). These changes of substrate elevation are not evenly distributed spatially, which may lead to different biases in mangrove TH growth estimations over a large study area. Therefore, it is necessary to separate the actual elevation change from the mangrove TH growth. Now with the highly accurate RTK-GPS function more conveniently integrated in UAV-based surveys, to measure the substrate elevation could be possible through UAV-LiDAR surveys. Hence, effort can be invested in monitoring TH change and substrate change jointly in the future.

5. Conclusion

This pioneering study marks the first attempt to characterize individual mangrove TH growth using multi-temporal UAV-LiDAR data. We demonstrated the feasibility of the method and proposed a UAV-LiDAR-based diagram for TH growth monitoring. With this diagram, we can quantify the TH growth rates of mangrove trees with considerably reduced human intervention, greatly enhancing the efficiency of mangrove growth monitoring and management. Both the grid-based and tree-based approaches could derive mangrove TH growth, with the tree-based approach showing a closer alignment with the field measured growth rates. Differences in the growth rates among different mangrove species were confirmed in this study, statistically significant differences among the clumpiness groups were discovered, while the impact of invasion on growth rate has not been found significant. It is acknowledged, however, that the findings are based on a small study area using a few different UAV-LiDAR sensors. To fully exploit the potential of multi-temporal UAV-LiDAR for tree dynamics monitoring, additional investigation into the algorithms, sensors, and environmental factors are indispensable. In addition, this study highlights the need of studies on the geo-registration of LiDAR data in forested areas.

CRediT authorship contribution statement

Dameng Yin: Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Le Wang:** Conceptualization, Data curation, Investigation, Methodology, Project administration, Resources, Writing – original draft, Writing – review & editing. **Ying Lu:** Data curation, Methodology, Writing – review & editing. **Chen Shi:** Methodology, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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